

Employee Ownership Support Programs and ESOP Participation

Evidence from Colorado's HB17-1214

Mitchell Ballif Patrick Beal Will Cheney Soren Pack

December 3, 2025

Did Colorado's policy increase ESOP adoption?

Why does this matter?

- ESOPs increase wealth for workers who own company shares
- Small businesses struggle with high transition costs
- Colorado offered education and financing in 2017
- Can this approach increase ESOP adoption?

Compare Colorado to Utah before and after 2017

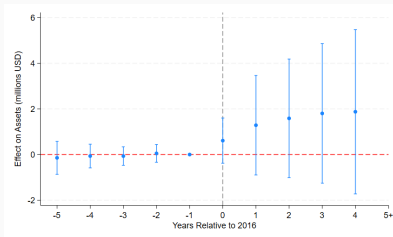
- Treatment: Colorado firms after 2017
- Control: Utah firms (similar economy, no policy)
- Data: 2012–2025 5500 Form pension filings
- State and year fixed effects
- Standard errors clustered at firm level

$$Y_{it} = \delta(\text{Colorado}_i \times \text{Post}_t) + \lambda_i + \mu_t + \epsilon_{it}$$

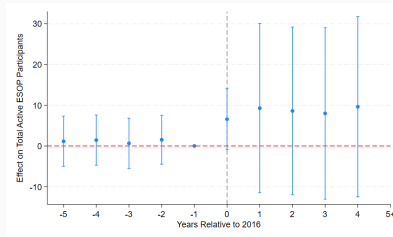
Where:

- Y_{it} : Outcome (ESOP firms, participants, assets)
- $\text{Colorado}_i \times \text{Post}_t$: Treatment indicator
- δ : DiD estimator (treatment effect)
- λ_i : State fixed effects
- μ_t : Year fixed effects

Testing Parallel Trends



Total ESOP Assets



Total ESOP Participants

- Before 2017: Colorado and Utah followed same trend
- Parallel trends assumption holds in the pre-trend

Are Our Assumptions Valid?

- ✓ **Compliance:** Treatment only in Colorado
 - HB17-1214 enacted exclusively in Colorado
 - Utah had no access to programs, no comparable policies
- ✓ **Parallel Trends:** Same trends before 2017
 - Event study shows pre-treatment coefficients ≈ 0
- ✓ **SUTVA:** No spillovers to Utah
 - Benefits require Colorado presence
 - ESOP transitions costly, unlikely cross-state influence
- ✓ **Exclusion Restriction:** Policy affects outcome only through ESOP adoption
 - HB17-1214 specifically targets ESOP formation
 - Education + loans directly reduce ESOP transition costs
 - Small budget (\$200k/year), no broader economic impacts
 - Policy doesn't affect firms through other channels

Main Results

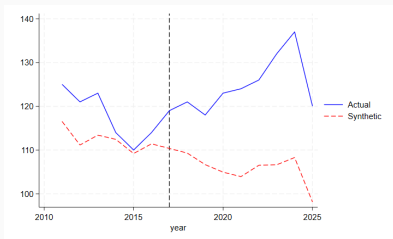
Outcome	Effect	SE
ESOP Firms	0.006	(0.005)
Total Participants	25.2	(17.4)
Total Assets (\$)	2.88M*	(1.61M)
Active Participants	22.3	(15.4)

The policy had no significant effect

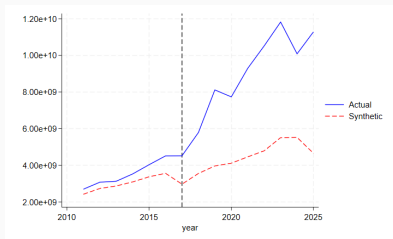
- Almost all results statistically insignificant
- Effects are economically small

MAJOR CAVEAT: Only 2 states $\Rightarrow$ cannot cluster at state level
Standard errors **understated**, precision **overstated**

Robustness: Synthetic Control



Number of ESOP Firms



Total ESOP Assets

- Synthetic control uses all US states as comparison
- Shows larger effects than our main analysis
- **Problem:** Poor fit before 2017 means results not reliable
- We trust our main DiD estimates more

Appendix: Full Regression Results

Table 1: *

Table 1: Difference-in-Differences Results (Year and State Fixed Effects)

Variable	(1) ESOP Firm	(2) Total Partcps.	(3) Total Assets	(4) Active Partcps.	(5) Partcp. Rate
DiD	0.00607 (0.00486)	25.19 (17.35)	2,878,102.7* (1,611,044.2)	22.29 (15.37)	0.00613 (0.00426)
Constant	0.0215*** (0.00131)	18.16*** (6.636)	1,258,818.5*** (445,403.5)	13.99*** (5.124)	0.0175*** (0.00113)
Observations	209,399	209,399	209,399	209,399	191,912

Standard errors (clustered at firm level) in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Note: Standard errors are clustered at the firm level. Theoretically appropriate state-level clustering is infeasible with only two states.